

DA Baseline Bias Ablation – S0/S1

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What this ablation does

Sweeps how badly DA baselines (Weak-4DVar, Strong-4DVar, EnKF, ETKF) degrade when the model they solve with is biased relative to the true Lorenz63 dynamics. Each baseline's Joint-* counterpart (Joint-Weak-4DVar, etc.) is swept alongside it -- same hyperparameters, but it also solves for sigma/rho/beta/c1 rather than using the (biased) DA-supplied values, so its state RMSE and its own parameter-recovery RMSE are both reported. Two independent bias mechanisms are swept, plus a joint sweep where both move together:

- * Parameter bias (param_bias)
biases the sigma/rho/beta values DA solves with; forcing_state_bias held at the scenario's own baseline
grid: [0, 0.05, 0.1, 0.15, 0.2, 0.3]
- * Forcing/state bias (forcing_state_bias)
biases the corrupted forcing fed to DA; param_bias held at the scenario's own baseline
grid: [0, 0.05, 0.1, 0.15, 0.2, 0.3]
- * Joint bias (joint_bias)
param_bias and forcing_state_bias vary together from 0 at matching values
grid: [0, 0.05, 0.1, 0.15, 0.2, 0.3]

Unlike the default S0/S1 baselines (evaluation/run.py::run_and_cache_baselines, which always feeds DA the TRUE forcing), this ablation feeds DA the CORRUPTED forcing, so forcing_state_bias actually perturbs what DA sees. Truth trajectories are unaffected by either bias – only the DA-facing sigma/rho/beta and corrupted forcing are biased – so varying bias does not change the physical windows.

Coupling exponent: a 4th, UNSWEPT bias fixed at every grid point

Truth couples W_L into dX nonlinearly, $c1 \cdot \text{sign}(W_L) \cdot |W_L|^{1.6}$ (coupling_exponent truth); DA -- vanilla AND Joint alike -- always solves with the LINEAR form $c1 \cdot W_L$ (coupling_exponent_da=1, never estimated, so Joint's sigma/rho/beta/c1 recovery can't fix it). Same gap for S0 and S1, all axes -- so the RMSE floor even at bias=0 already reflects this fixed model-form error. (Unrelated to forcing_coupling 'linear'/'quartic' in conf/schema.py, used only to train the NN surrogates elsewhere in this repo, not this DA ablation.)

Scenarios

- S0: perfect operating point (no bias) (param_bias=0.0, forcing_state_bias=0.0, num_windows=50)
- S1: biased operating point (param_bias=0.15, forcing_state_bias=0.1, num_windows=50)

Methods

- | | |
|--------------------|--|
| Weak-4DVar | opt_steps=150, lr=0.02 |
| Joint-Weak-4DVar | opt_steps=150, lr=0.02 (+ jointly estimates sigma/rho/beta/c1) |
| Strong-4DVar | max_iter=40, lr=0.1 |
| Joint-Strong-4DVar | max_iter=40, lr=0.1 (+ jointly estimates sigma/rho/beta/c1) |
| EnKF | N_ensemble=30, inflation=2.0 |
| Joint-EnKF | N_ensemble=30, inflation=2.0 (+ jointly estimates sigma/rho/beta/c1) |
| ETKF | N_ensemble=30, inflation=2.0 |
| Joint-ETKF | N_ensemble=30, inflation=2.0 (+ jointly estimates sigma/rho/beta/c1) |

Source: experiments/ablation_bias_s0_s1.json
Runs found: 36

Sweep Definitions — What Each Axis Actually Varies

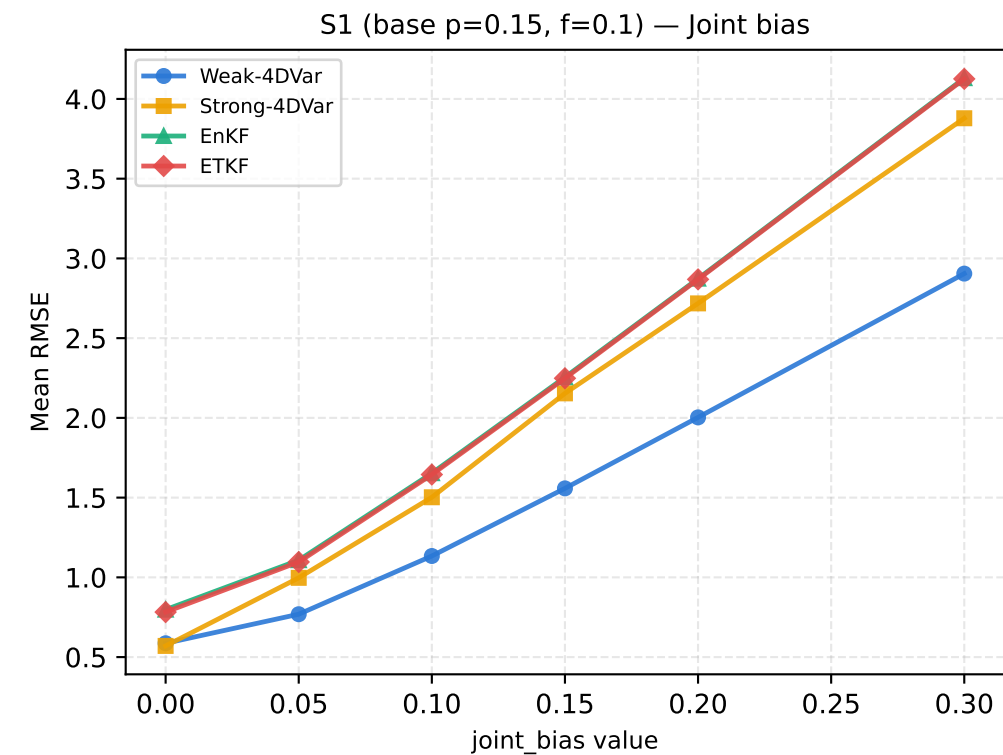
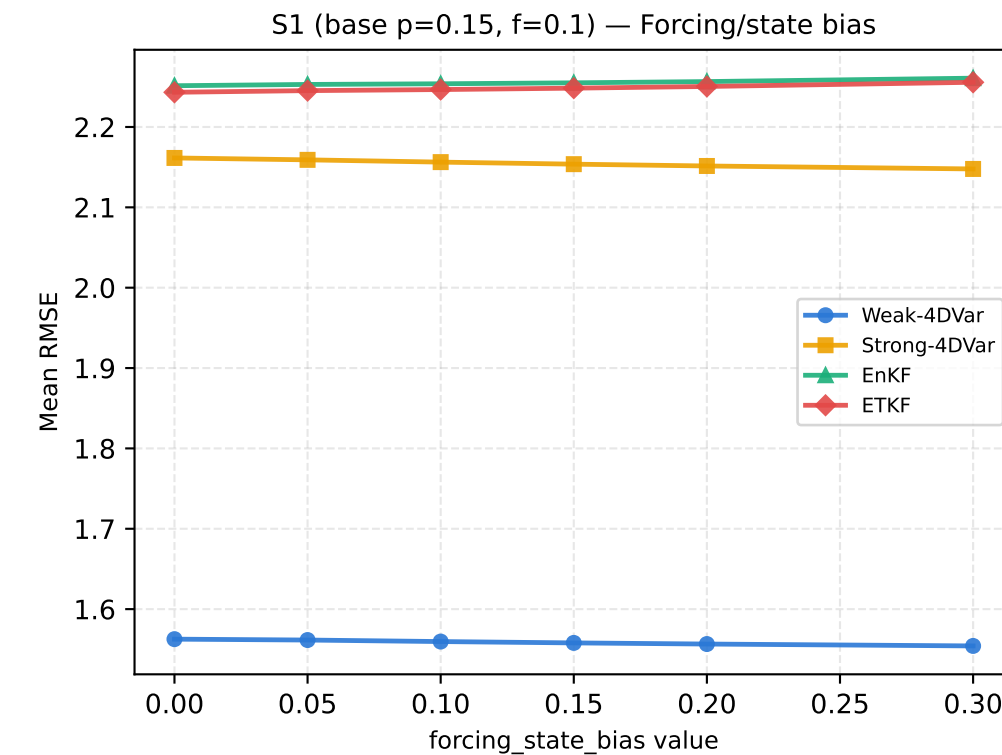
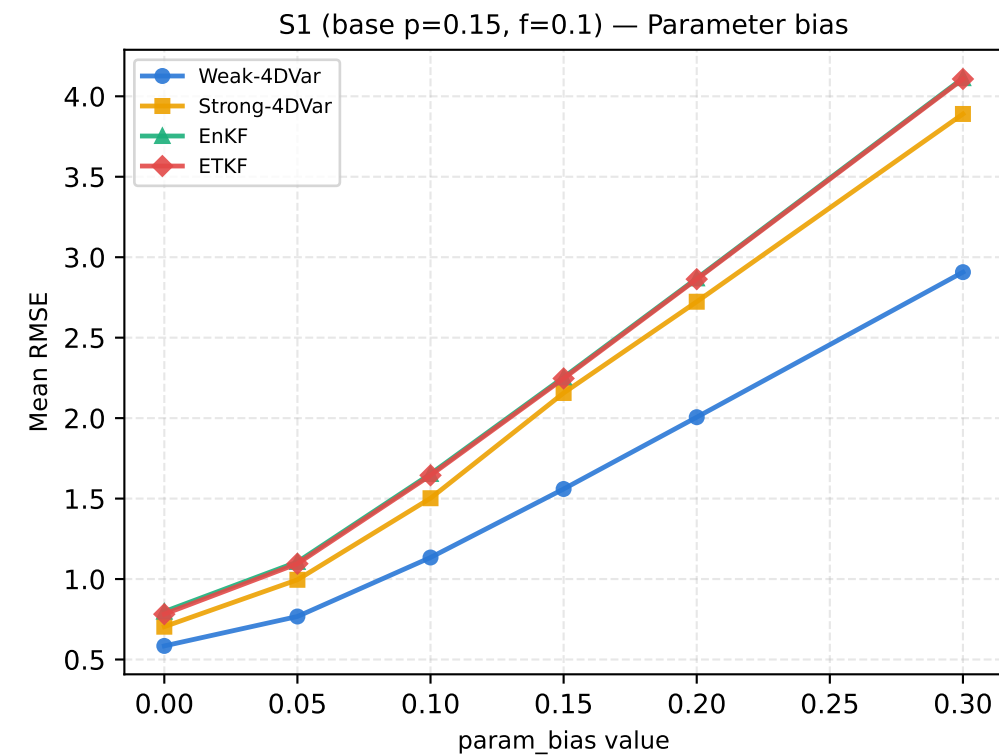
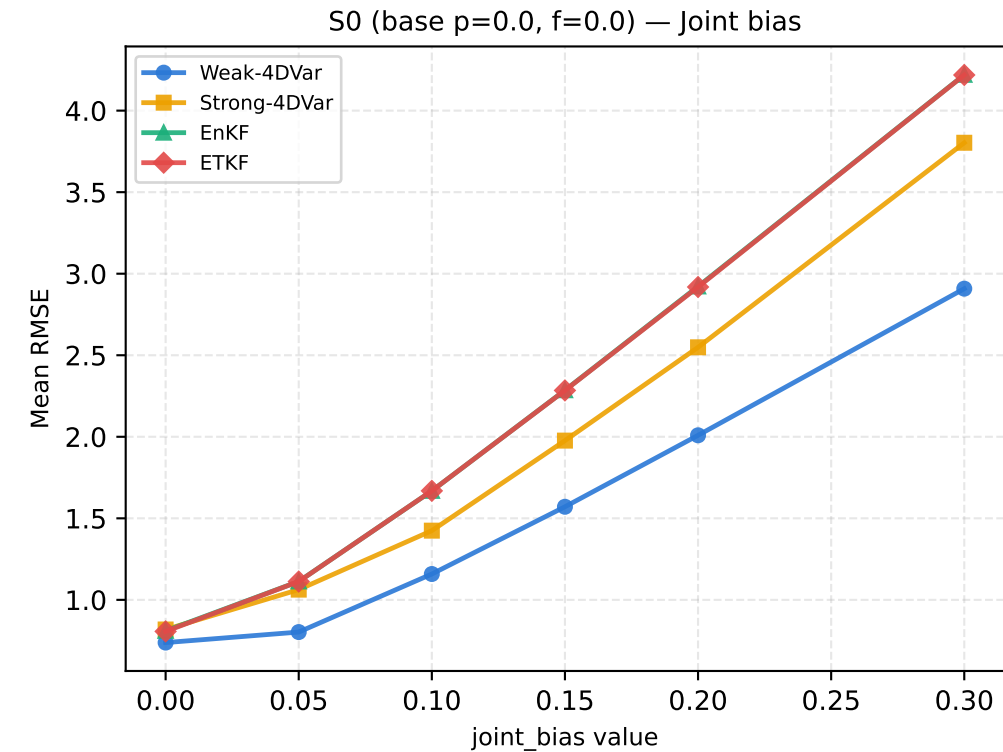
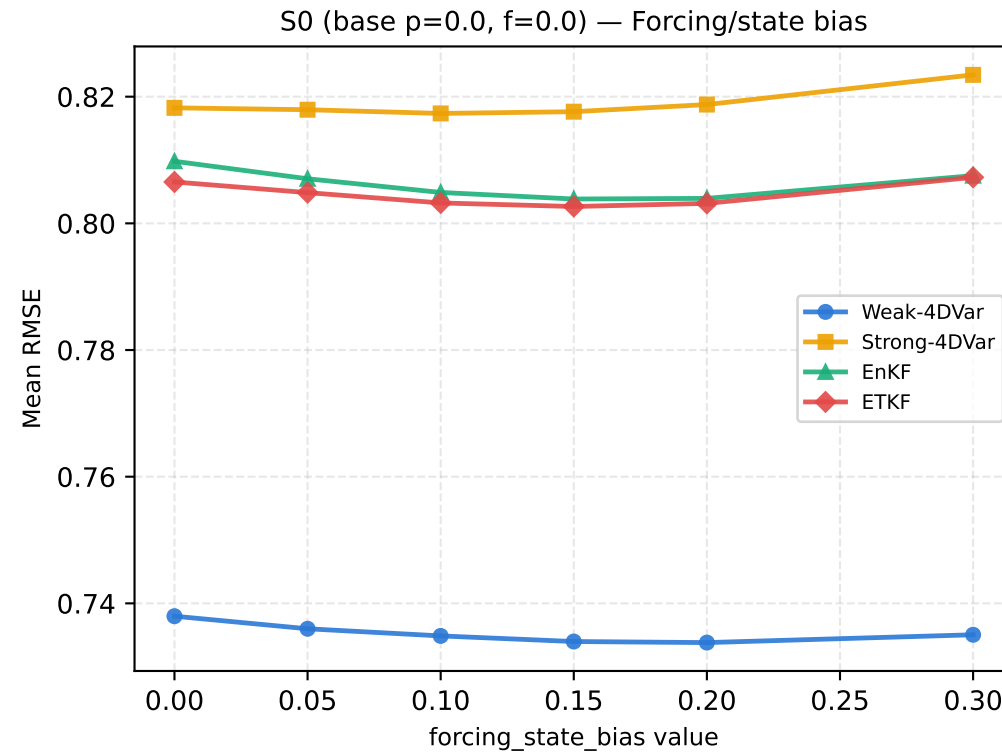
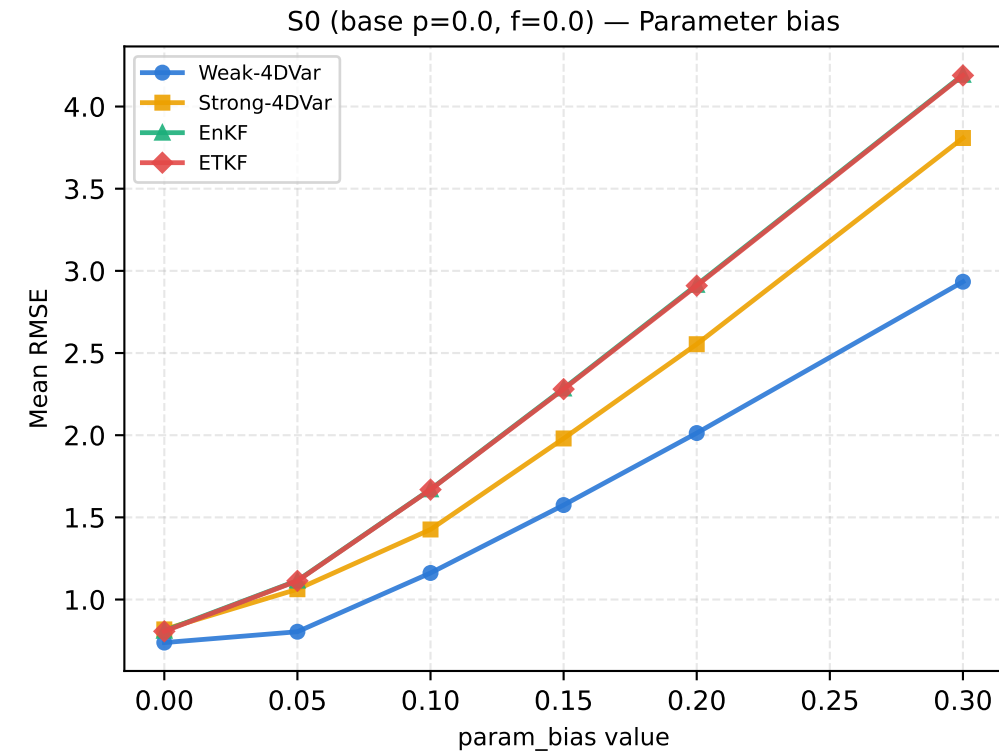
S0 sweeps (anchored at param_bias=0.0, forcing_state_bias=0.0)

axis	param_bias	forcing_state_bias
param_bias sweep	0, 0.05, 0.10, 0.15, 0.20, 0.30	fixed at 0.0
forcing_state_bias sweep	fixed at 0.0	0, 0.05, 0.10, 0.15, 0.20, 0.30
joint_bias sweep	0, 0.05, 0.10, 0.15, 0.20, 0.30	same value (0, 0.05, 0.10, 0.15, 0.20, 0.30)

S1 sweeps (anchored at param_bias=0.15, forcing_state_bias=0.1)

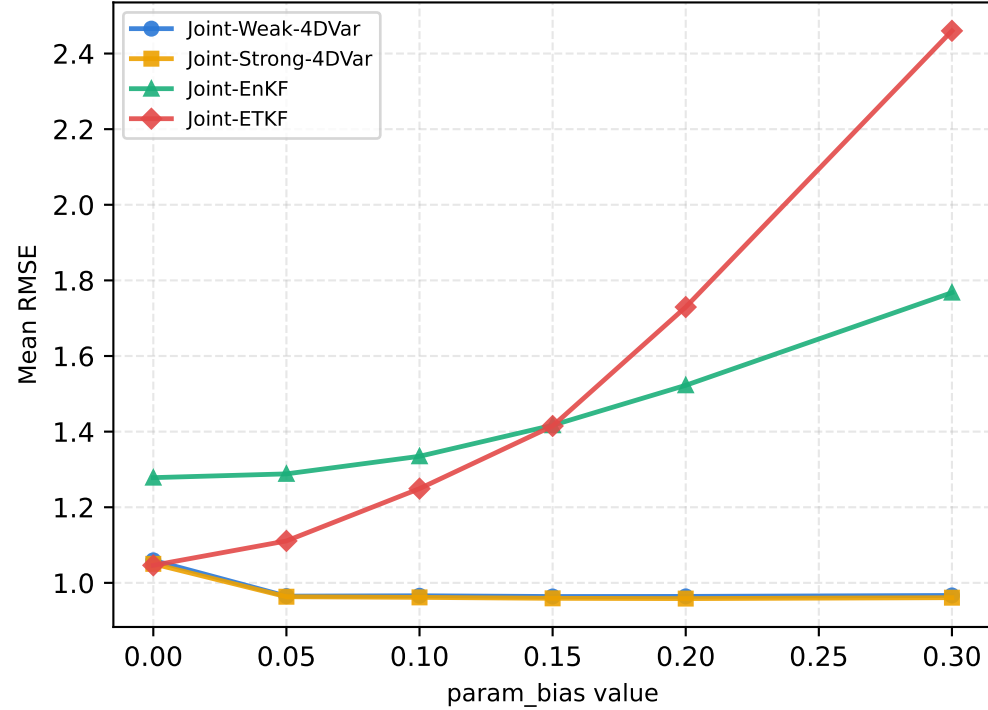
axis	param_bias	forcing_state_bias
param_bias sweep	0, 0.05, 0.10, 0.15, 0.20, 0.30	fixed at 0.1
forcing_state_bias sweep	fixed at 0.15	0, 0.05, 0.10, 0.15, 0.20, 0.30
joint_bias sweep	0, 0.05, 0.10, 0.15, 0.20, 0.30	same value (0, 0.05, 0.10, 0.15, 0.20, 0.30) — note: joint sweep starts both at 0, not anchored to S1's 0.15/0.1 base

Default Methods — Mean State RMSE vs Bias Magnitude

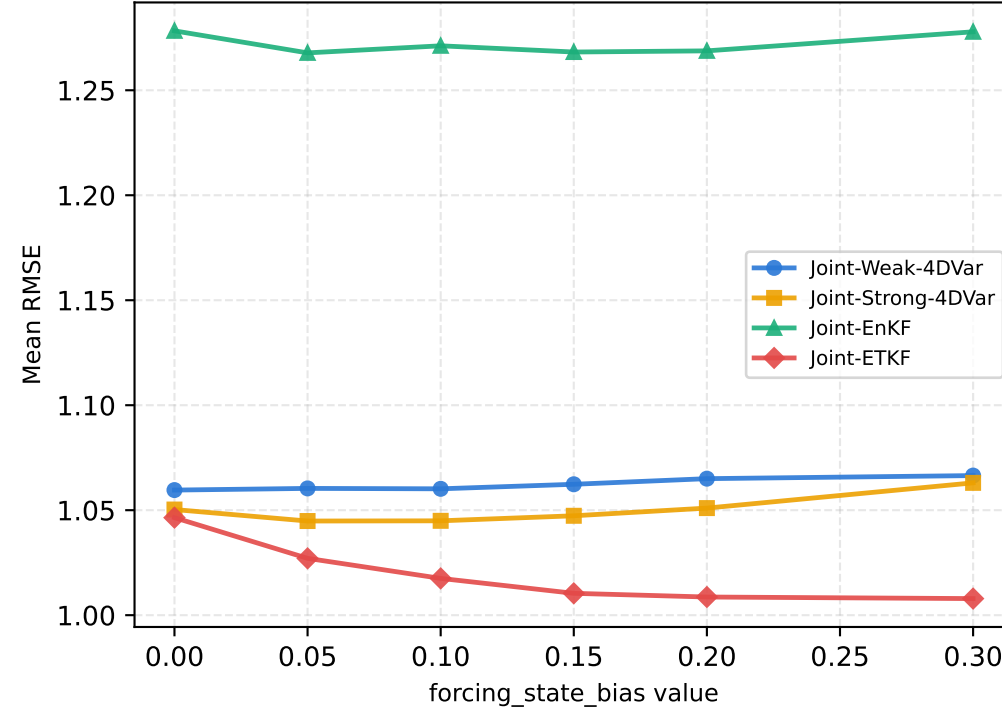


Joint Methods — Mean State RMSE vs Bias Magnitude

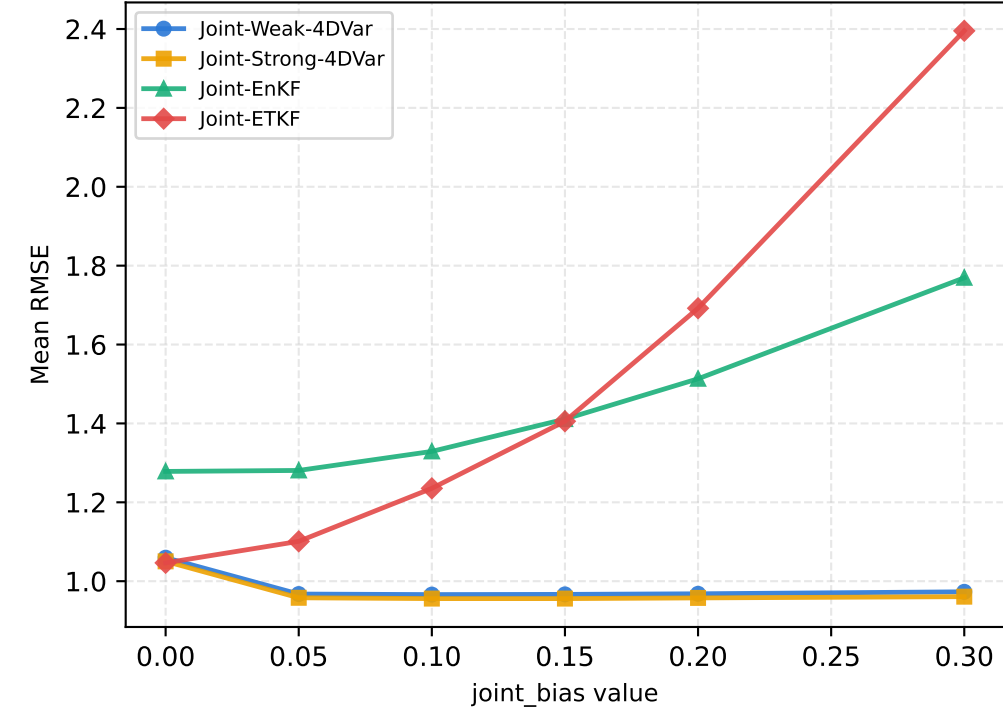
S0 (base p=0.0, f=0.0) — Parameter bias



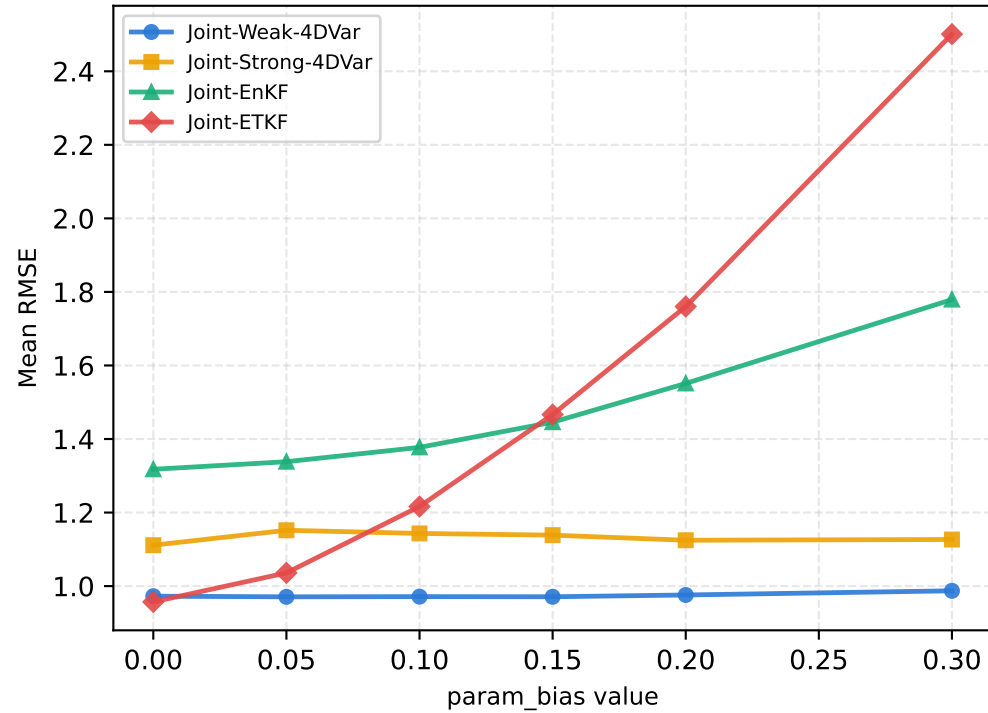
S0 (base p=0.0, f=0.0) — Forcing/state bias



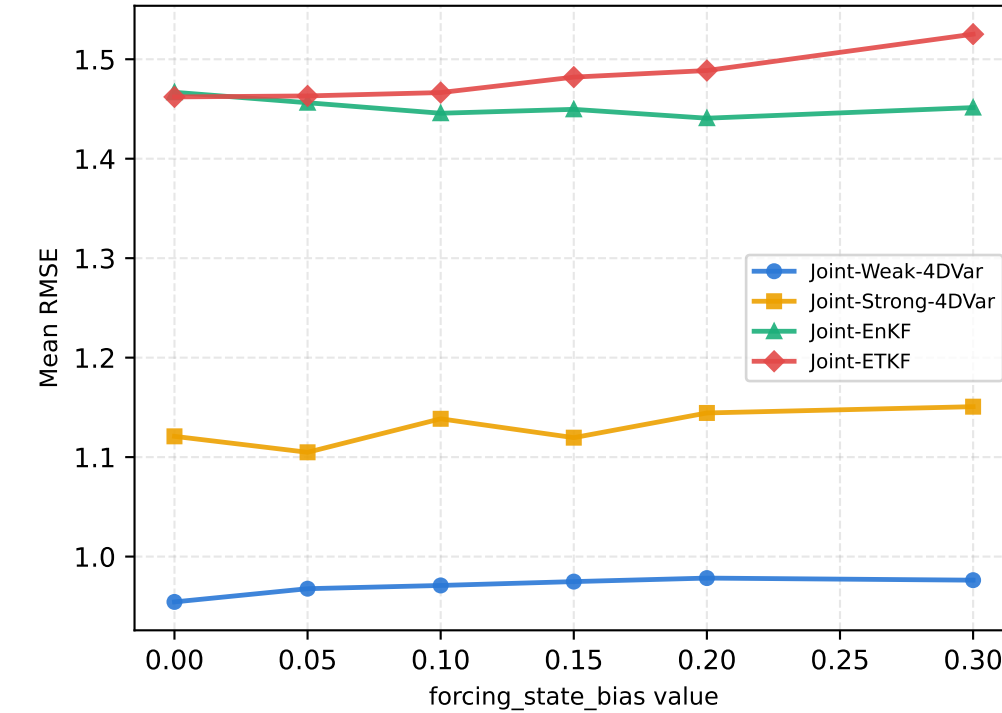
S0 (base p=0.0, f=0.0) — Joint bias



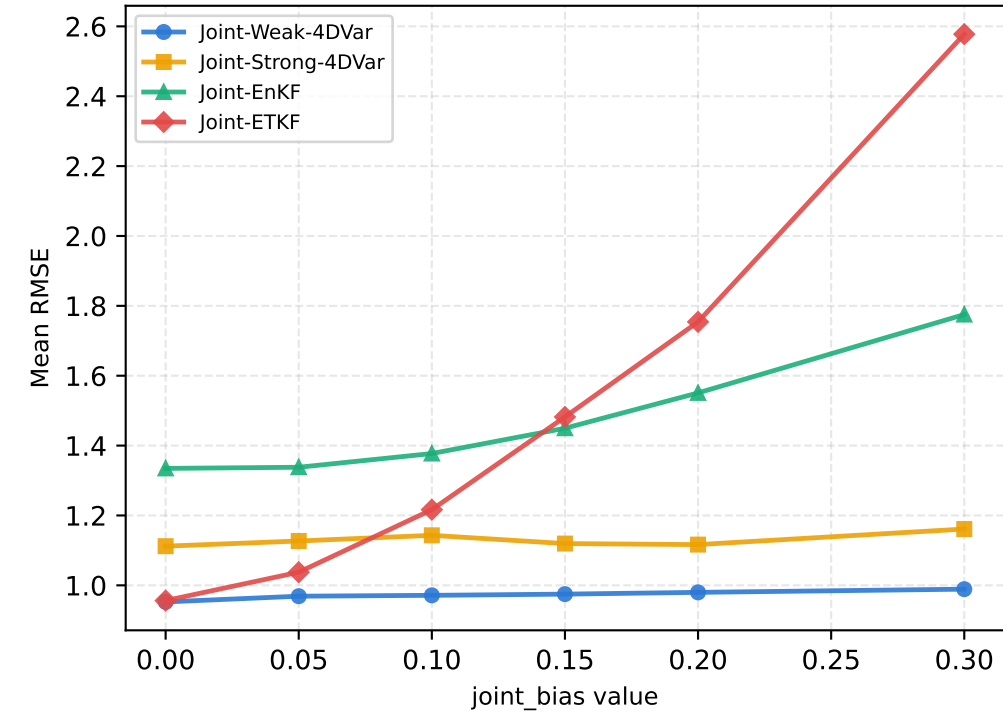
S1 (base p=0.15, f=0.1) — Parameter bias



S1 (base p=0.15, f=0.1) — Forcing/state bias



S1 (base p=0.15, f=0.1) — Joint bias



Joint Methods — Mean Parameter RMSE vs Bias Magnitude (avg of sigma/rho/beta/c1)

